AI-Based Personalized Course Recommendation and Learning Path System

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Department: Computer Science and Engineering

Subject: Foundations of Artificial Intelligence

Domain/Algorithm: A* Search

Abstract

PathFinder is a personalized learning-path application that helps a student plan courses toward a selected career goal. It compares existing skills with the target role's required skills, identifies gaps, and searches for a prerequisite-valid course sequence.

INPUT

- Current skills and knowledge level
- Target career goal
- Study availability and interests

AI METHOD

- A* state-space search
- g(n): cost accumulated so far
- h(n): estimated remaining cost

OUTPUT

- Ordered course sequence
- Skill-gap summary and cost
- Study plan and progress views

Problem Statement & Objectives

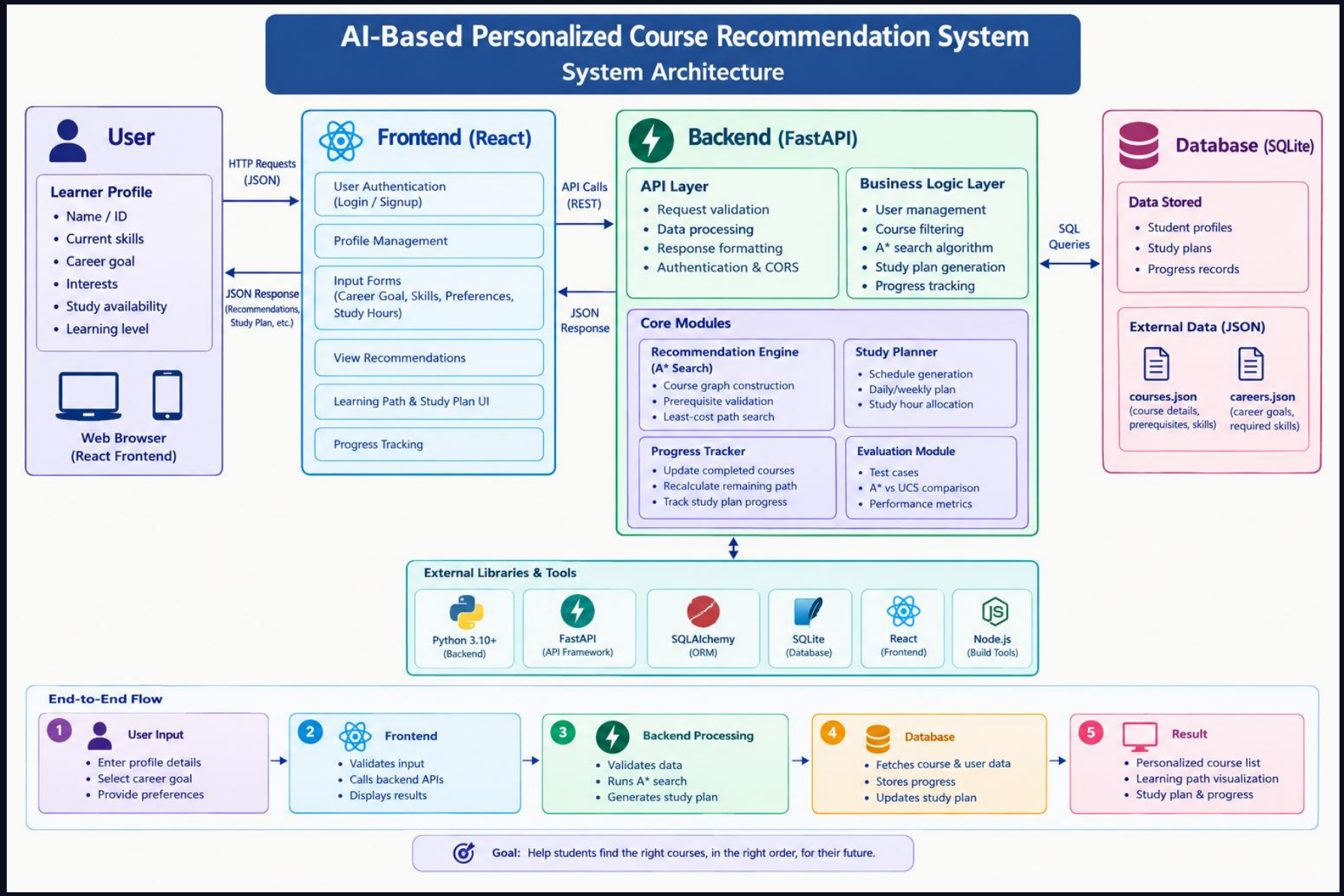
PROBLEM STATEMENT

- Students may not know which skills are missing for a chosen career.
- Courses taken in the wrong order can cause prerequisite gaps.
- Generic course lists do not account for existing skills or study capacity.
- The project generates a goal-directed sequence from a defined course catalogue using A* search.

OBJECTIVES

- Collect student profile, current skills and career goal.
- Compare acquired skills with career requirements.
- Respect prerequisites when generating a path.
- Estimate search cost using course duration and difficulty penalties.
- Present results, timetable and progress tracking.

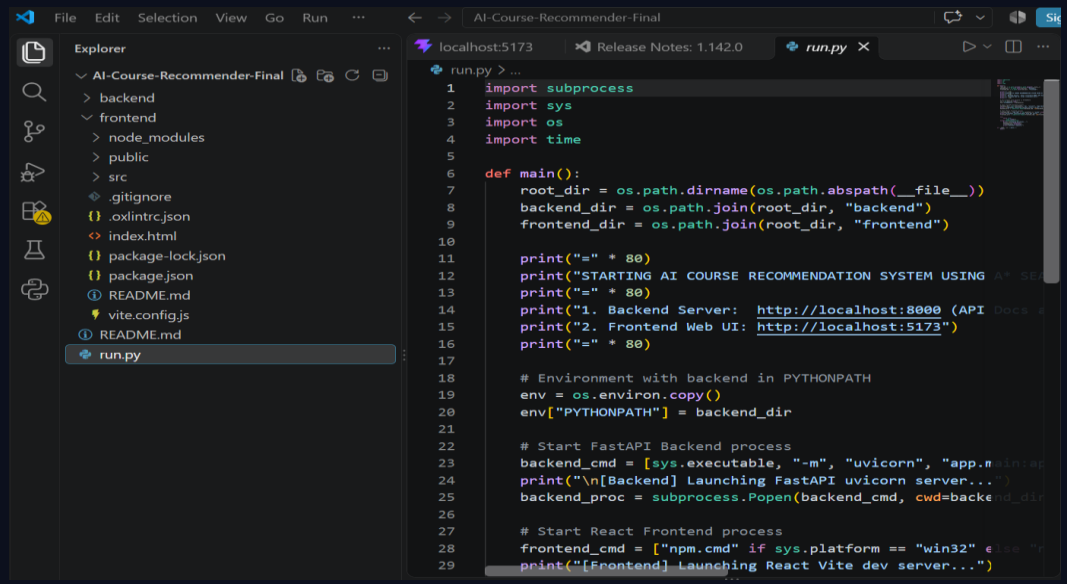
System Architecture Diagram

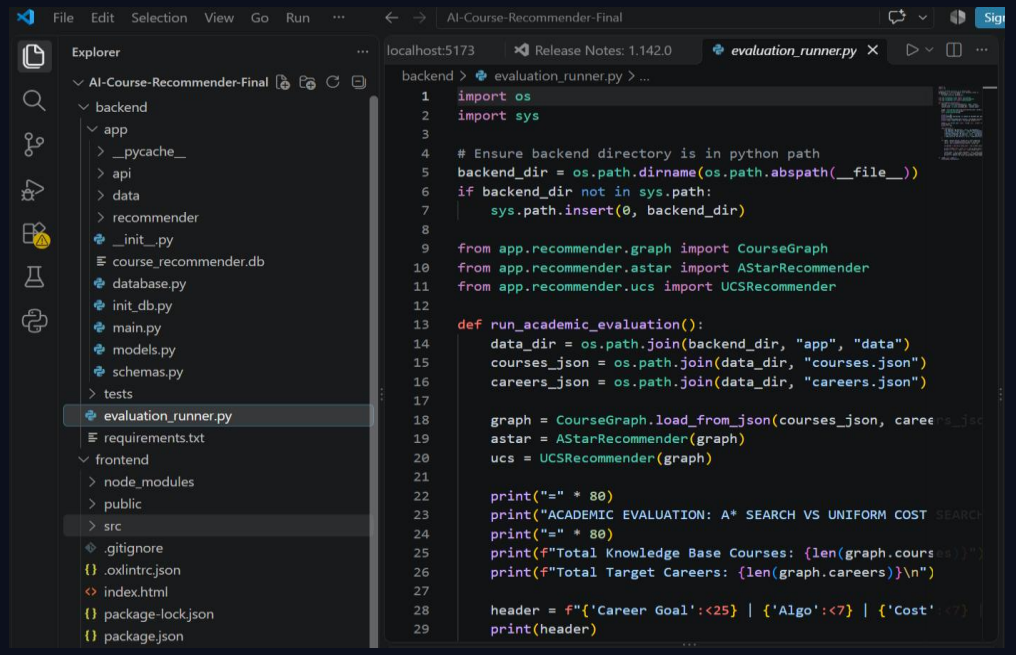


A* Search: Algorithm & Sample Code

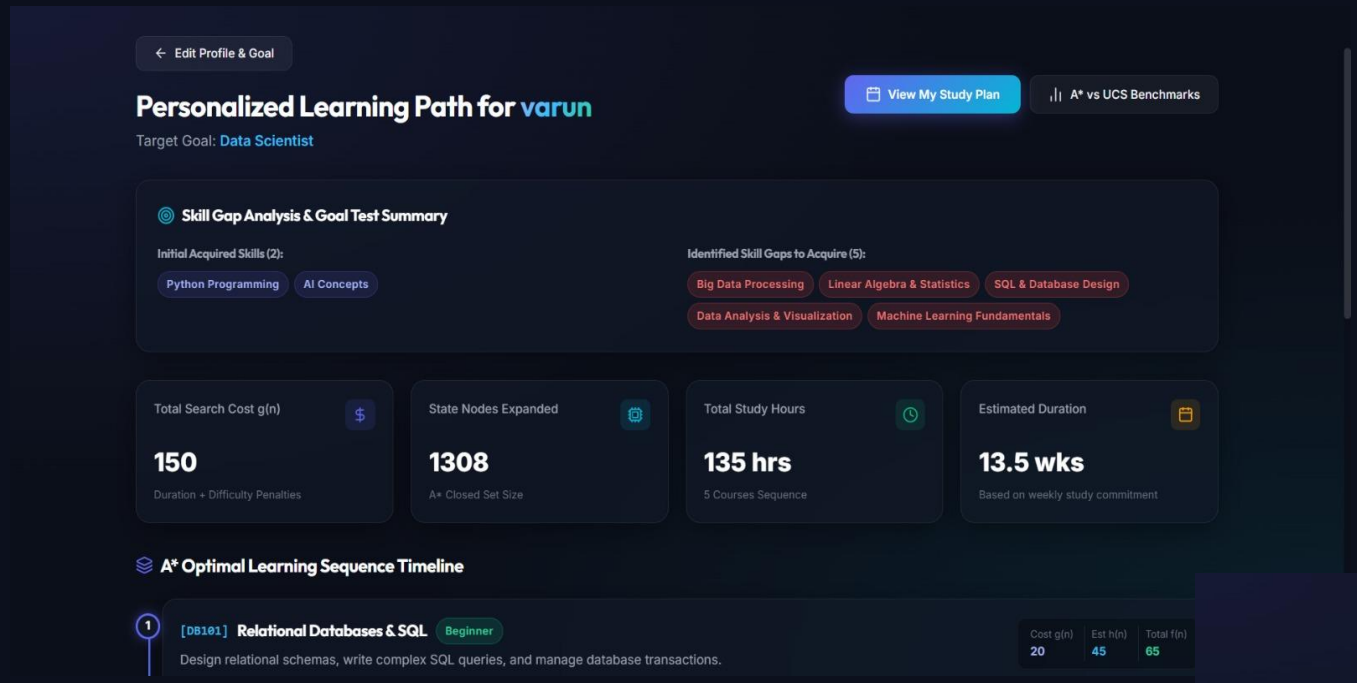
A* FORMULATION

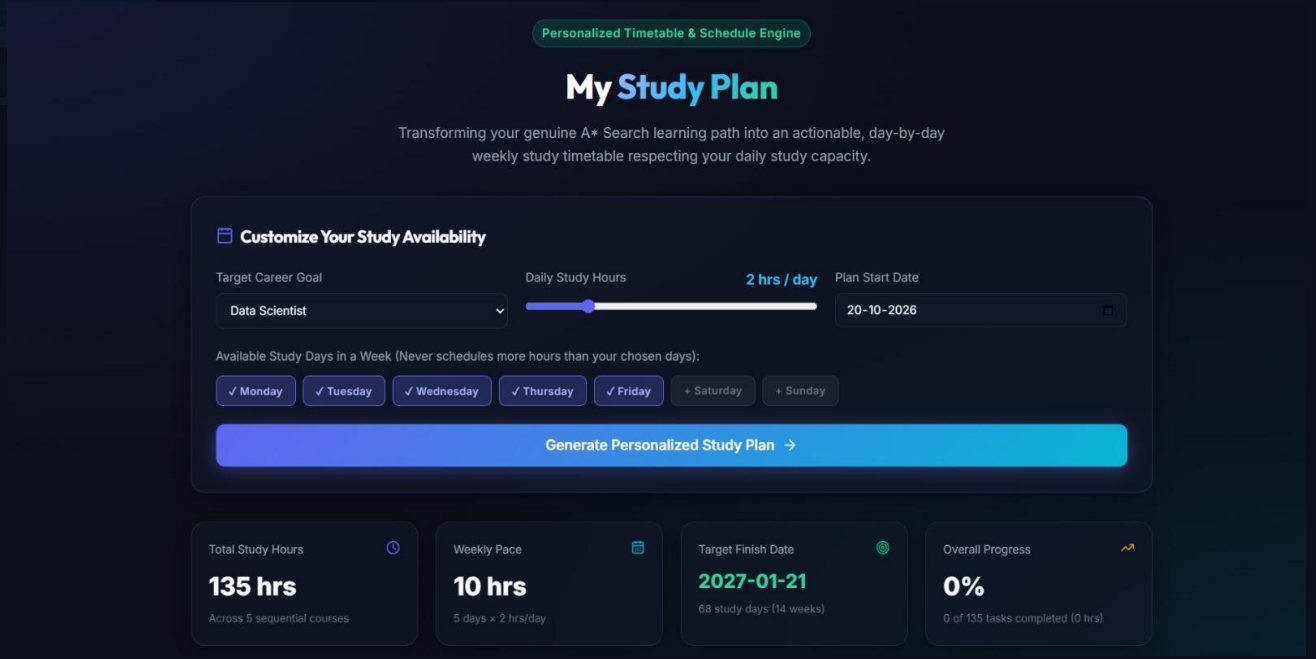
- State: completed skills/courses.
- Successor: course whose prerequisites are satisfied.
- g(n): accumulated cost of selected courses.
- h(n): estimate of remaining goal cost.
- f(n) = g(n) + h(n): expansion priority.
- Goal test: all required career skills are covered.





Sample Output





Conclusion

PathFinder A* demonstrates how heuristic search can support personalized educational planning. By combining a learner profile, skill-gap analysis, course prerequisites and cost-aware search, the system presents an ordered learning path through a dashboard.

KEY TAKEAWAY

- Career goal → skill gaps → course path

AI CONCEPT

- A* evaluates f(n) = g(n) + h(n)

PRACTICAL VALUE

- Study plan and progress-aware replanning

References

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• Python Software Foundation. Python 3 Documentation. https://docs.python.org/3/

• Project implementation and UI screenshots supplied by the project author.